

Basic QM:

1. The orthonormal eigenfunctions $\psi_n(\vec{r})$ of a Hermitian Operator Q form a complete set. Write an arbitrary normalised wave function $\psi_n(\vec{r}, t)$ using $\psi_n(\vec{r})$.
2. For a system described by a time independent potential, the stationary state solutions are given by $\psi_E(\vec{r})$, where E is the energy eigenvalue. Write the general solution $\psi_E(\vec{r}, t)$.
3. For a free particle, find the eigenfunction of the momentum operator and how that its wavelength equal the De Broglie wavelength.
4. If the eigenfunctions and eigenvalues of the operator $\hat{A}$ are ϕ_n and a_n respectively, and the function $f(x) = \sum_{i=0}^{\infty} c_i \phi_i$. Show that ϕ_n is eigenfunction of $f(\hat{A})$ with value $f(a_n)$ ($f(\hat{A})\phi_n = f(a_n)\phi_n$).
5. Write the expression for the rate of change of the expectation value of an operator Q (the wave functions satisfy the time dependent Schroedinger equation). If the expression holds for any arbitrary function e.g. $\langle \Phi | Q | \Psi \rangle$, then write the corresponding operator equation.
6. Using the commutator algebra for $\vec{L}$, show that $\hat{L}^2$ commutes the three components of $\vec{L}$.
7. Show that

$$\int_{-1}^{+1} P_l(\cos\theta) P_{l'}(\cos\theta) d(\cos\theta) = \frac{2}{2l+1} \delta_{ll'} \quad (1)$$
8. Plot the probability distribution $|Y_{lm}(\theta, \phi)|^2$ for $Y_{0,0}$, $Y_{1,\pm 1}$, $Y_{2,0}$, and $Y_{2,\pm 1}$.
9. For a given value of the orbital and spin angular momenta l and s , list the possible value of j and m_j where $\vec{J} = \vec{L} + \vec{S}$.
10. For a central force potential, write the time-independent H operator in spherical polar coordinate and $\hat{L}^2$ operator. Show that $[H, \hat{L}^2] = [H, L_z] = 0$.
11. Find the Eigenvalue of the parity operator $\hat{P}$. Write the function $\psi(\vec{r})$ in terms of parity eigenfunctions.
12. For a system of N particles with individual momenta $\vec{L}$ and $\vec{S}$, write the commutator relation for $\vec{r}_i$ and $\vec{p}_j$.
13. Write the eigenvalues of the permutation operator $\hat{P}_{ij}$.
Note: In general $[\hat{P}_{ij}, \hat{P}_{ik}] \neq 0$ and so, the eigenfunctions $\psi(q_1, q_2, \dots, q_n)$ of H are *in general* not eigenfunction of $N!$ permutation/exchange operator *except* ψ is *totally symmetric or anti-symmetric*.

14. Show that

$$\int_{-\infty}^{+\infty} F(t, \omega) d\omega = t \int_{-\infty}^{+\infty} \frac{\sin^2 x}{x^2} dx = \pi t \quad (2)$$

$$\text{Where, } F(t, \omega) = \frac{1 - \cos(\omega t)}{\omega^2} \quad (3)$$

15. Show that $\hat{L}^2$, $\hat{S}^2$, $\hat{J}^2$, and $\hat{J}_z$ form a commuting set of operators and $[\hat{L}, \hat{S}, \hat{J}_z] = 0$

16. Consider a particle of charge q and mass m which is in simple harmonic motion along the X axis with force constant k . An electric field $E(t)$, directed along the x-axis, is switched on at time $t=0$ so that the system is perturbed by an interaction $H'(t) = -qx E(t)$.

If $E(t)$ has the form $E(t) = E_0 e^{-t/\tau}$ where E_0 and τ are constants. If the oscillator is in ground state for $t \leq 0$, Find the probability that the oscillator will be found in an excited state as $t \rightarrow \infty$.